

Before taking the big jump from IBM 1130 class machines to Burroughs B 6700s, these universities conducted a thorough—and revealing—comparison of computers from several manufacturers

Evaluating Computers for the New Zealand Universities

by John Good
and
B. A. M. Moon

In 1970 the New Zealand Universities were faced jointly with choosing new computing equipment to provide for most of them a quantum jump in computer power and range of facilities. This paper describes the authors' techniques, first to establish present and future power requirements of the universities in a principally FORTRAN-based environment and second to measure the extent to which these requirements were likely to be met by the different manufacturers' proposals.

As can be seen from Map 1, New Zealand has six universities plus an agricultural college of university status. These are scattered over the length of both North and South Islands, spanning a total distance of some 800 miles. They are autonomous and primarily state-supported with some supplementary capital and revenue from endowments and grants made by the private sector, with additional revenue provided by student fees. State finance for the universities is allocated in advance for five-year periods and disbursed through the University Grants Committee. In February, 1970, the New Zealand government made known that it was prepared to fund capital expenditure on computing equipment during the 1970 to 1974 quinquennium. Universities were requested jointly to seek and consider alternative re-equipment proposals and to make recommendations to government through the University Grants Committee. The universities were happy enough to concur with this plan for, though all possessed some sort of

computing equipment already (see Map 1), they were all restricted by the very limited range of facilities provided. Universities in the North Island were already experiencing saturation of their facilities while those in the South Island expected to reach saturation by 1972.

Map 1 also shows the final decision announced by the government in August, 1971, to install Burroughs B6700 computers at the five larger institutions and to serve each of the smallest with a data communications link to the adjacent B6700 site.

After detailed consideration of a large number of alternative proposals from six computer companies this was felt to be the best way to meet the needs of all the universities for access to systems providing a wide variety of programming languages, operating primarily in a batch-processing mode with some time-sharing capacity.

As part of the selection process, it was necessary to establish the computer and peripheral powers needed by the universities and to measure these aspects of the vendors' proposals. This paper describes some of the techniques used by the authors in this work. While it was carried out on behalf of the New Zealand Universities Vice Chancellors' Committee, the opinions expressed here are those of the authors.

Given certain basic capabilities (in this case local and remote batch modes with some time-sharing), at any given time a computing facility must provide certain basic levels of computing power, printing capacity, and card reader

speed. If these are not provided then some of the work cannot be done. These are quantitative factors and therefore measurable on proposed equipment. As needs they are assessable and, within limits, predictable.

There are also qualitative factors, such as the range of programming languages, the flexibility of the operating system and the quality of printed output which are of greater or lesser importance in assessing the merit of a proposal and which can only be given subjective valuations.

Many factors—for example, core size and backing store capacity and speed—have both qualitative and quantitative aspects.

Our objective was to establish some of the more fundamental quantitative factors.

Estimating quantitative needs

To establish the quantitative needs required us to determine the load in a base year and estimate the growth rate. The latter is necessarily predictive but past experience offered a useful guide.

During the eight years of computing experience at Canterbury University, when disruptive factors have been absent, computer use has doubled every two years and, on grounds of continuity, the same type of basic growth rate is expected in the future. However, the acquisition of new equipment is itself a disruptive factor. It may reduce the work load, because the jobs will be done more efficiently; it may increase it because the new facilities will not only satisfy immediately visible needs

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Another disruptive factor experienced at both Canterbury and Otago was the introduction of charging for computer time. A charging system for use of the Canterbury 360/44 was instigated in 1969 and the immediate

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To determine the base load, we studied the pattern of computer load distribution at Canterbury University and the composition of the various institutions in the country. It was necessary to establish the relative per capita use between graduate and undergraduate

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1969 - 540 Students

IBM 1130

1972 - DC1200 Terminal

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1971 - IBM 1130

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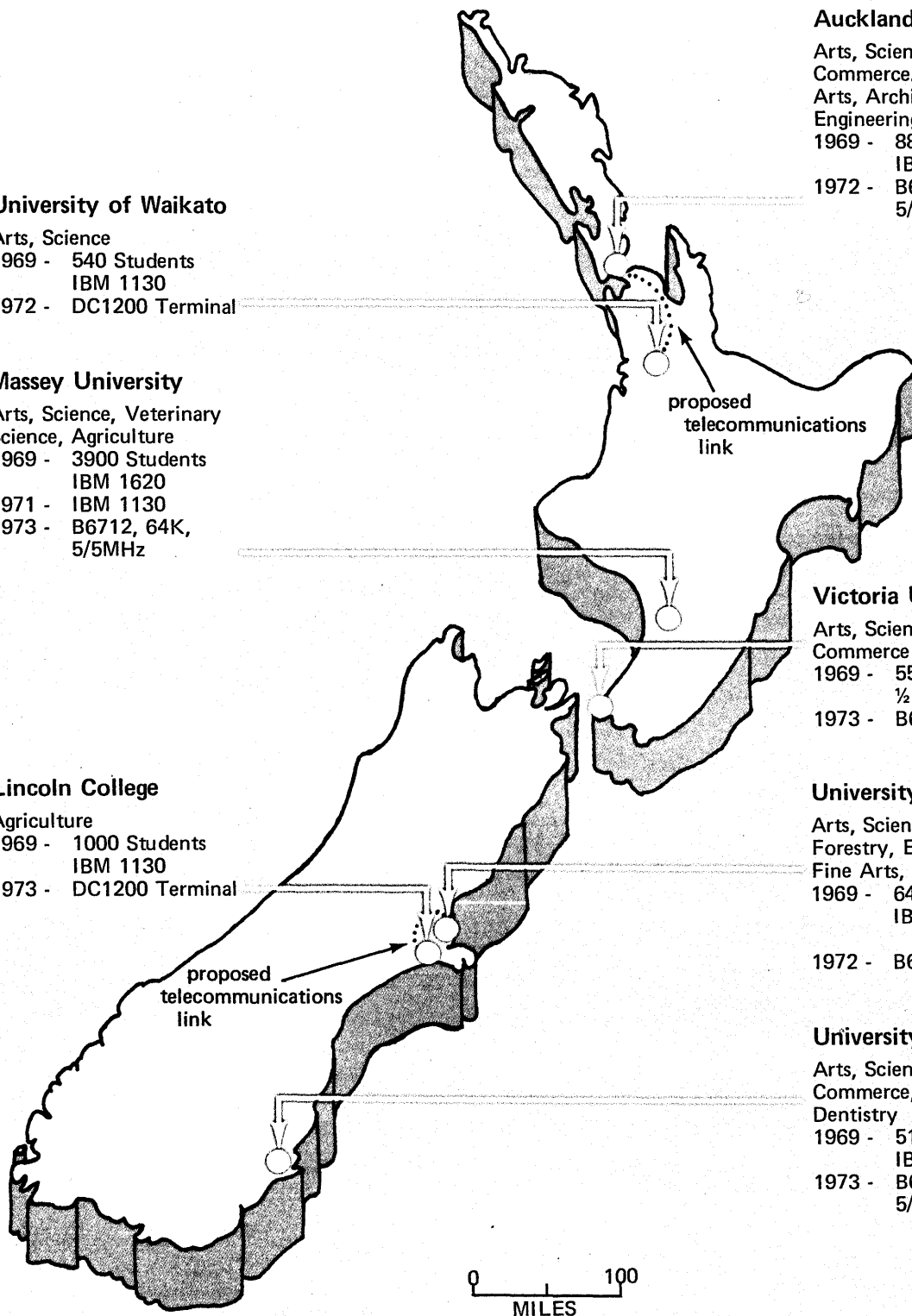
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Faculty members themselves contribute appreciably to computer use but this is closely related to student load in most faculties and so treating the load as if it all came from students was sufficient for predictive purposes. The faculties whose distribution of use between staff and students is atypical (e.g., medicine) contribute only a

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Available evidence showed that the relative per capita load graduate/undergraduate was likely to be around 40:1 and that the total picture was not sensitive to variations in this ratio.

Equating forty undergraduates to one graduate the number of equivalent graduate students (EGS) per faculty was obtained. Dividing the number of hours of annual computer use by the number of EGS for each faculty gave annual per capita values. These were expressed as average rates of annual usage of a fully utilized 360/44 cpu,

which was a convenient unit of power for subsequent benchmarking purposes.

Similar data were available from some other universities and, after making the necessary adjustments to reflect the much lesser power of their computing equipment, it was possible to incorporate their experiences into an overall predicted need for the New Zealand Universities. Actual average hours of 360/44 cpu use per EGS for major contributing faculties and their share of the overall total on a national basis at the 1969 level are given in Table 1.

Application to the loads per faculty of the basic growth rate, planned changes in relative faculty sizes and disruptive factor estimates yielded an estimated power requirement for each faculty at the end of the period to be served by the proposed new equipment. Summing for all the faculties,

	EGS	HOURS/EGS	SHARE OF TOTAL
Sciences	962	2.71	41%
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	B6700	80K,1.2usec (5/5MHz)	MCP	511	644	79
CDC	6400	64K	SCOPE	199	240	83
	3300	Not known	MASTER		1,440 ¹	
IBM	360/44	128Kbytes	44PS	1,120	4,459	25
			44MFT		2,100 ³	53 ³
	360/50	512Kbytes	OS/MVT	998	1,163	86
	360/65	1024Kbytes	OS/MVT	298	370	81
	360/67	1024Kbytes	MTS	426 ²	453	94
	360/75	1024Kbytes	OS/MVT	207	255	81
	370/135	240Kbytes	OS/MVT	1,050*	1,500*	70*
	370/145	572Kbytes	OS/MVT	525*	656*	80*
ICL	1904S	128K	GEORGE3		780 ¹	
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³ Improved 360/44 throughput obtained with a slightly modified benchmark under 44MFT, use of CANTRAN in-core compiler and a spooling system.

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MACHINE	IN TERMS OF A 360/44 CPU ⁽³⁾	TO A SCALE OF 19
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CDC 6400	4.67	19
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IBM 370/155*	3.62	15
IBM 360/65	3.03	12
ICL 1906A ⁽²⁾	2.73	11
IBM 360/67	2.47	10
DEC PDP-10/I ⁽²⁾	2.29	9
Univac 1106	2.04	8
Burroughs B6700 5/5MHz	1.74	7
IBM 370/145*	1.71	7
ICL 1904S ⁽²⁾	1.44	6
(360/44 CPU)	(1.00)	4
IBM 360/50	0.96	4
DEC PDP-10/A ⁽¹⁾	0.90	4
CDC 3300	0.78	3
IBM 370/135*	0.75	3
Burroughs B5500	0.68	3
IBM 360/44 best throughput	0.53	2
Original 360/44 throughput	0.25	1

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Benchmark programs

The tool used for measurement of cpu power needs and performances was a batch of some 34 FORTRAN programs. Individually some of the programs investigated particular features of the FORTRAN language and hardware of the configuration under test. Collectively they formed a model of some of the individual characteristics, like core occupancy and distribution of elapsed times, of the University of Canterbury's job stream. Most programs were "live" ones written by ordinary computer users at Canterbury, Auckland and Otago Universities as part of their personal studies. A realistic proportion contained known "bugs."

The batch of programs was known to be compiler-intensive in comparison with the job stream. It was also input-output intensive. This was revealed by running it as a datum on the 360/44 at Canterbury University and timing the activities of various components of the system with an IBM 2989 Basic Counter Unit (BCU). This showed that the batch was 24.6% cpu active while the job stream which it was modeling was probably about 35% cpu active. Given the total number of cards and lines of print involved in the batch it was possible to obtain an indication of the rated speeds of card readers and printers suited to the cpu speeds given in the different proposals for the New Zealand university environment, after making some allowance for the stimuli towards both greater efficiency and also greater use which more advanced systems were likely to have on patterns of input-output. However, the major purpose of the exercise was to provide a base level of performance (the cpu time recorded on the BCU for the batch of programs when run under the 44 Programming System on the 360/44) against which the throughput power of each configuration under test could be measured.

A model of a FORTRAN jobstream was the only benchmark of this type practicable for the New Zealand Universities and certainly this technique could be criticized for narrowness. However, as a measure of overall system performance we believe that it gives a sufficiently accurate measure of quantitative performance for comparative purposes, especially since, no matter what other languages are provided, the present investment in FORTRAN programs and expertise and the relative absence of programs and expertise in other languages at most New Zealand Universities will make FORTRAN likely to be the most used language for some years ahead. Were a deficiency to be

discovered in a particular compiler, other than FORTRAN, the New Zealand Universities might contemplate making the necessary software improvements themselves. If a machine is inadequate in FORTRAN for the New Zealand Universities' environment it is fundamentally inadequate for their purposes.

Machine performance

Almost all machines of interest to New Zealand Universities perform their input-output operations with hardware independent from the cpu. There may be competition for memory; only in one or two cases are cpu cycles "stolen." It very quickly became clear that adequate input-output capacity was attainable on all the machines to be tested. Attention was therefore focused upon the ability of the cpu/FORTRAN compiler/operating system/core/backing store combination to execute the benchmark. Two items were of interest.

The first was the disc-to-disc (in some cases, magnetic tape-to-magnetic tape) elapsed time required for execution of the benchmark. The second was the cpu time required to execute the benchmark programs, excluding cpu time spent on system overheads except on the datum 360/44 whose cpu overheads were minimal. It could broadly be said that the cpu time thus recorded was a measure of the raw power of the compiler/cpu/core speed combination while the disc-to-disc elapsed time reflected how much of this power was permitted to be used by the operating system/backing store/core size combination.

Table 2 shows the cpu and disc-to-disc elapsed times used. It also shows cpu time expressed as a percentage utilization of elapsed time. These figures are given for the different configurations tested, together with estimates for certain machines which were not available for testing in 1970. Almost all the times are to some extent estimated because it was not possible to obtain identical output from the different companies. One lesson learned in our analysis was the importance of giving precise unambiguous instructions to those responsible for running the benchmark. They should be given a copy of the exact program execution output expected. We did not do this and the results suffered accordingly.

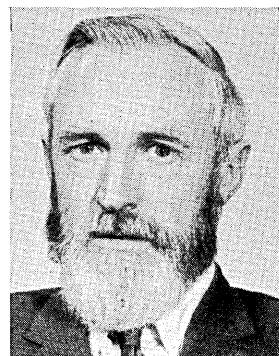
Table 3 shows the powers of the various configurations in terms of the throughput powers of the 360/44 cpu and also rounded into a scale of 19. It will be noticed that the Univac 1108 and CDC 6400, which were the most powerful machines tested, have throughput capacity 19 times that of the original 360/44 configuration and that the 18 computers investigated are spread quite evenly across that range.

Readers should be clear that these performance figures refer to execution in a batch environment of a model of the University of Canterbury's 360/44 job stream. The figures serve as a valuable pointer to the likely power of each machine configuration in the general New Zealand university environment. Extension of these figures to cover environments substantially involving other methods of use and different programming languages would need some caution. Far better that readers investigate their past and present use, ascertain their likely growth rates and construct their own benchmark model to reflect their usage.

Every acquisition of new computing equipment involves a leap into the unknown and the leap was a very large one for the New Zealand Universities. This paper describes a method designed to wring the last ounce of useful information out of past and present use so that future use could be more accurately predicted and most economically provided for. □



John Good is chief programmer/analyst at the computer center of the Univ. of Canterbury, New Zealand. He previously worked for a computer manufacturer in Johannesburg, South Africa, and at a commercial installation in Auckland, New Zealand. He has B.Sc. degrees from Leeds Univ., England.



Bruce Moon is director of the computer center at the Univ. of Canterbury. He has worked with computers for the Australian government and for IBM World Trade Corp. He has M.Sc. degrees from the Univ. of New Zealand in mathematics and the Univ. of London in computer science. He is also a Fellow of the Institute of Physics, London.

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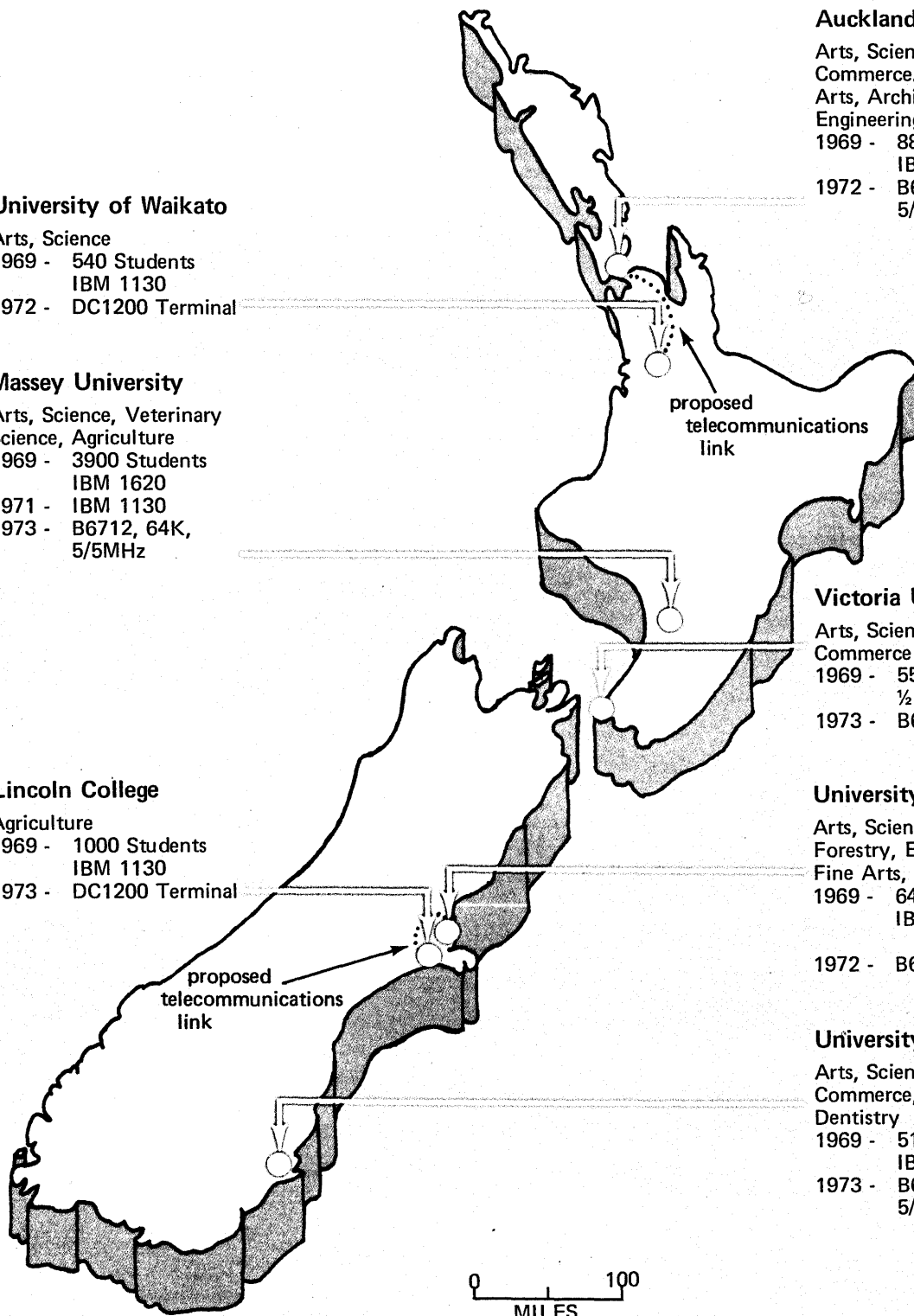
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Machine performance

Almost all machines of interest to New Zealand Universities perform their input-output operations with hardware independent from the cpu. There may be competition for memory; only in one or two cases are cpu cycles "stolen." It very quickly became clear that adequate input-output capacity was attainable on all the machines to be tested. Attention was therefore focused upon the ability of the cpu/FORTRAN compiler/operating system/core/backing store combination to execute the benchmark. Two items were of interest.

The first was the disc-to-disc (in some cases, magnetic tape-to-magnetic tape) elapsed time required for execution of the benchmark. The second was the cpu time required to execute the benchmark programs, excluding cpu time spent on system overheads except on the datum 360/44 whose cpu overheads were minimal. It could broadly be said that the cpu time thus recorded was a measure of the raw power of the compiler/cpu/core speed combination while the disc-to-disc elapsed time reflected how much of this power was permitted to be used by the operating system/backing store/core size combination.

Table 2 shows the cpu and disc-to-disc elapsed times used. It also shows cpu time expressed as a percentage utilization of elapsed time. These figures are given for the different configurations tested, together with estimates for certain machines which were not available for testing in 1970. Almost all the times are to some extent estimated because it was not possible to obtain identical output from the different companies. One lesson learned in our analysis was the importance of giving precise unambiguous instructions to those responsible for running the benchmark. They should be given a copy of the exact program execution output expected. We did not do this and the results suffered accordingly.

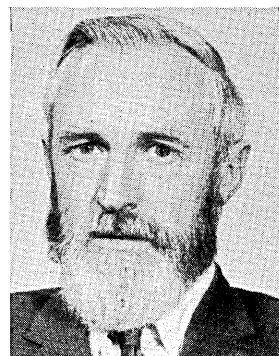
Table 3 shows the powers of the various configurations in terms of the throughput powers of the 360/44 cpu and also rounded into a scale of 19. It will be noticed that the Univac 1108 and CDC 6400, which were the most powerful machines tested, have throughput capacity 19 times that of the original 360/44 configuration and that the 18 computers investigated are spread quite evenly across that range.

Readers should be clear that these performance figures refer to execution in a batch environment of a model of the University of Canterbury's 360/44 job stream. The figures serve as a valuable pointer to the likely power of each machine configuration in the general New Zealand university environment. Extension of these figures to cover environments substantially involving other methods of use and different programming languages would need some caution. Far better that readers investigate their past and present use, ascertain their likely growth rates and construct their own benchmark model to reflect their usage.

Every acquisition of new computing equipment involves a leap into the unknown and the leap was a very large one for the New Zealand Universities. This paper describes a method designed to wring the last ounce of useful information out of past and present use so that future use could be more accurately predicted and most economically provided for. □



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